



Course Title: Advanced Technologies and Clinical Decisions in Acute Care

Course Number: NURS 7640

Semester: Fall 2025

Credit: 1.0 credit unit

Total hours of instruction: 45 hours

Catalog Description: This course applies a technology assessment framework to examine numerous advanced technologies employed in the care of acutely or critically ill and/or chronically complex patients. Students investigate how technologies are used to assess, diagnose, manage and/or evaluate the health status of patients. Students enhance their decision-making regarding the benefits and limitations of each technology through the use of evidence-based research. Students are introduced to the psychomotor skills used to perform select invasive procedures. Adult-Gerontology ACNP and Pediatric ACNP students participate in shared content and program specific content.

Canvas Page: <https://canvas.upenn.edu/courses/1862476>

Faculty, Contact Information, and Office Hours:

Patricia Griffith, PhD, CRNP-BC (Course Director)

pgriffit@nursing.upenn.edu, 267-252-1049 mobile

Office Hours: By appointment

Ng (Tony) Niu, MSN, CRNP-BC (Associate Course Director)

Kerry Shields, MSN, MBE, CRNP (Pediatric Concentration Director)

Jessica Strohm Farber, DNP, CRNP (Pediatric Acute Care Program Director)

Prerequisite Courses:

AGACNP Students: N6070 (prerequisite); N6570, N6700 (pre/co-requisite); N6730 (co-requisite)

Peds ACNP students: N6850 and N7210 (co-requisites)

*All students must have a minimum one-year of nursing experience before taking this course

Objectives: At the end of this course, the student will be able to:

1. Demonstrate proficiency in advanced assessment and use of technologically derived data to inform advanced nursing practice of acutely ill or injured and/or chronically complex patients.
2. Develop clinical judgment in the use of diagnostic and/or therapeutic technologies for individuals and populations.
3. Use scientific inquiry to examine best practices for advanced technologies in managing



acutely ill or injured and/or chronically complex individuals and populations.

4. Explore the informed consent process including patient and family education when preparing to apply or employ diagnostic and/or therapeutic technologies.
5. Engage in socially just, ethical, and cultural practices for the delivery of advanced nursing care related to diagnostic and/or therapeutic technologies.
6. Apply skills for collaboration and leadership to direct care for acutely ill or injured and/or chronically complex patients.

Teaching Methods:

Didactic content delivery is provided by experienced clinicians' presentations, readings, online videos, and websites. Level II competencies are developed through lab practicums, deliverables, and quizzes, and are demonstrated through summative exams and a course project. This course uses Canvas to organize content, readings, assessments, and communication.

All **communication** will be conducted through your school of nursing email. Clear, respectful, and prompt communication is an expectation of our profession. We as faculty will respond to any individual student email within 48 hours; we expect, and require, that students will respond to a course/faculty email within 48 hours. Failure to do so could result in missed critical course information; continued communication failures may result in program dismissal.

Class attendance is a crucial element in your nurse practitioner education. The exchange of ideas and ability to ask questions to the expert lecturers will help contribute to your success in this course. Nurse Practitioner certification is contingent upon a specific number of clinical and theory class hours. Upon graduation, the program director must sign certification paperwork that confirms that each student has received the minimum number of course hours. *For this reason, attendance will be taken at each class.*

Any anticipated absence needs to be cleared by Patti/Kerry (program dependent) prior to the absence. In the event of an illness or family emergency, please email or text Patti/Kerry by the end of business on that class day. An absence will be considered "unexcused" if you have not contacted Patti/Kerry within that period. An unexcused absence will require a meeting with Patti. Two "unexcused" absences or three "excused" absences will require a meeting with your program director and may result in a reduction of your class grade by 2 points per absence. Course content will be recorded for viewing and available for viewing later the same day; virtual attendance to an in person class is not permitted.

Statement of Welcome, Respect, Belonging, and Community:

Penn Nursing demands that all persons experience a learning environment that celebrates us all as individuals and as a community, respecting our identities and diversity. Dimensions of diversity include but are not limited to sex, race, age, national origin, immigration status, ethnicity, gender identity and expression, intellectual and physical abilities, sexual orientation,



income, faith and spirituality, socio-economic class, political ideology, education, primary language, family status, military experience, cognitive style, and communication style. We expect every member of our classroom community to contribute to an inclusive and respectful culture. The individual intersection of experiences and characteristics are valued in our classroom. We support every student in every classroom, whether in-person or on a virtual platform, to find and embrace the ability to speak up and participate. Because we thrive on the diversity of everyone's beliefs, backgrounds, and experiences, we expect that each member of this class is ensured respect, fairness, and equality consistent with our mission, values, and guiding principles. In that way, we affirm that we are an inclusive community who value our universality and our diversity. As such, we are curious, empathic, and supportive of one another, of our fellow people, and of the planet. We hold awareness of ourselves and other beings as central to our work as nurses. We acknowledge our limitations and seek to expand on them, taking responsibility when we misstep or make mistakes. With active hope, we join with one another in learning, in care, and in commitment.

Class rosters are provided to instructors with the student's legal name. Our instructors will gladly honor your request to address you by any preferred name/pronouns/honorifics. Please advise your instructors of your request early in the semester so that they may make appropriate changes to language used in the classroom.

Academic Integrity:

Academic Integrity is fundamental to this course. [The International Center for Academic Integrity](#) states that 'Academic integrity is a commitment, even in the face of adversity, to six fundamental values: honesty, trust, fairness, respect, responsibility, and courage. From these values flow principles of behavior that enable academic communities to translate ideals into action.' Adherence to the [University of Pennsylvania Code of Academic Integrity](#) is required of every student in all coursework and all course interactions. Please refer to Penn Nursing guidance on [University of Pennsylvania Student Policies](#).

Course Grading Scale and Passing Grade Policy:

A+ 97-100	B+ 87-89	C+ 77-79	F 0-69
A 93-96	B 83-86	C 73-76	
A- 90-92	B- 80-82	C- 70-72	

A minimum of a C- is required to pass all graduate level courses. A grade lower than this will be reported as an F on the student's transcript.

Final course letter grade computation: All course grades will be rounded to the nearest percentage point when assigning the final letter grade. Example: 96.5 becomes 97/A+; 96.49 remains a 96/A

Evaluation Methods:



Each evaluation method is listed as an assignment on Canvas where it is further detailed including the due date and assessment rubric

● Technology Presentation Description	25%
● Tests (2).....	40%
● Quizzes (3).....	20%
● Student Created Multiple Choice Questions & Peer Review (AGACNP).....	10%
● Unfolding case studies (PACNP).....	10%
● Consent/procedure notes.....	5%
● Speaker Feedback (weekly).....	P/F

Required Readings:

Topic specific readings/videos/websites are found under each corresponding Module on Canvas

Textbooks with library links:

Irwin, R. S., Lilly, Mayo, & Rippe, J. M. (2024). Irwin & Rippe's Intensive Care Medicine 9e. Wolters Kluwer [Library Link](#)

Marino, P. (2025). The ICU book. (5th ed). Philadelphia: Lippincott, Williams & Wilkins. ISBN-13:978-1-9751-7670-9. [Library link](#)

Recommended Texts/Additional Resources:

Recommended topic specific readings/videos/websites are found under the corresponding Canvas module

Course Calendar:

The programs share content where applicable; population-specific content is also included in this course. For this reason, each program has its own calendar.

AGACNP:

PedsACNP: